FLIGHT BOOKINGS & TRAVEL DATA ANALYSIS

General Information :

1. Number of Rows: 250
2. Number of Columns: 9

Purpose:

- Simulated dataset representing airline ticket bookings, useful for data analysis, visualization and business insights in the aviation/travel industry.

Value Distributions

1. Airlines: Randomly chosen among 6 international airlines.
2. Travel_Class: Balanced mix of Economy, Premium Economy, Business, and First.
3. Departure_City: Covers major global hubs (London, Mumbai, Dubai, New York, Singapore, Frankfurt, Delhi, Paris).
4. Payment_Method: Multiple payment modes included for diversity.

Possible Use Cases

1. Exploratory Data Analysis (EDA): Ticket price trends, passenger age distribution, duration vs distance analysis.
2. Visualization: Compare average ticket prices by airline, travel class, or city.
3. Business Insights: Identify most common payment methods, popular routes, and preferred travel classes.

Some possible questions to solve with this dataset:

1. Which airline has the highest average ticket price?
2. Is there a correlation between distance and ticket price?
3. What is the average passenger age for each travel class?
4. Which departure city shows the longest average flight duration?
5. Which payment method is most popular for business-class travelers?
6. Do older passengers prefer certain airlines or payment methods?

Exploratory Data Analysis

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv(r'C:\Users\Srikant Verma\Desktop\Data Analytics\Power
        BI\Colortech Training\Flight Bookings & Travel Data
        Analysis\Project Files\Flight Booking & Durations.csv')
```

```
df.head()
```

	Booking_ID	Ticket_Price	Distance_km	Flight_Duration_hr	Passenger_
0	FB1001	3015.91	12243.3	23.92	60
1	FB1002	416.23	1034.1	1.42	52
2	FB1003	2897.73	7842.7	12.66	59
3	FB1004	4173.46	1987.1	2.22	38
4	FB1005	2500.49	14944.2	25.75	65

```
df.tail()
```

	Booking_ID	Ticket_Price	Distance_km	Flight_Duration_hr	Passenge
245	FB1246	766.65	6273.8	9.63	24
246	FB1247	3278.84	10887.8	15.77	37
247	FB1248	678.12	14396.2	28.32	30
248	FB1249	4878.30	8615.7	13.52	65
249	FB1250	3165.45	7601.5	10.98	54

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Booking_ID          250 non-null    object
 1   Ticket_Price        250 non-null    float64
 2   Distance_km         250 non-null    float64
 3   Flight_Duration_hr  250 non-null    float64
 4   Passenger_Age       250 non-null    int64
 5   Airline             250 non-null    object
 6   Travel_Class        250 non-null    object
 7   Departure_City      250 non-null    object
 8   Payment_Method      250 non-null    object
dtypes: float64(3), int64(1), object(5)
memory usage: 17.7+ KB
```

```
df.describe()
```

	Ticket_Price	Distance_km	Flight_Duration_hr	Passenger_Age
count	250.000000	250.000000	250.000000	250.000000
mean	2578.259040	7556.954000	11.041600	45.596000
std	1500.475213	4161.211576	6.394044	20.724643
min	113.350000	220.500000	0.260000	13.000000
25%	1203.885000	4211.775000	5.695000	28.000000
50%	2604.485000	7658.600000	11.220000	43.000000
75%	3986.327500	11369.950000	15.882500	62.000000
max	4985.360000	14944.200000	28.320000	80.000000

```
df.isnull().sum()
```

```
Booking_ID            0
Ticket_Price          0
Distance_km           0
Flight_Duration_hr    0
Passenger_Age         0
Airline               0
Travel_Class          0
Departure_City        0
Payment_Method        0
dtype: int64
```

KPI Creation

Total Bookings

```
Total_Bookings = df['Booking_ID'].nunique()

print(Total_Bookings)
```

```
250
```

Total Ticket Price

```
Total_Ticket_Price = df['Ticket_Price'].sum()

print(Total_Ticket_Price)
```

```
644564.76
```

Average Passenger Age

```
Average_Passenger_Age = df['Passenger_Age'].mean().round(2)

print(Average_Passenger_Age)
```

```
45.6
```

Total Distance

```
Total_Distance = df['Distance_km'].sum()

print(Total_Distance)
```

```
1889238.5
```

Average Flight Duration

```
Average_Flight_Duration = df['Flight_Duration_hr'].mean().round(2)

print(Average_Flight_Duration)
```

```
11.04
```

Average Ticket Price

```
Average_Ticket_Price = df['Ticket_Price'].mean().round(2)

print(Average_Ticket_Price)
```

```
2578.26
```

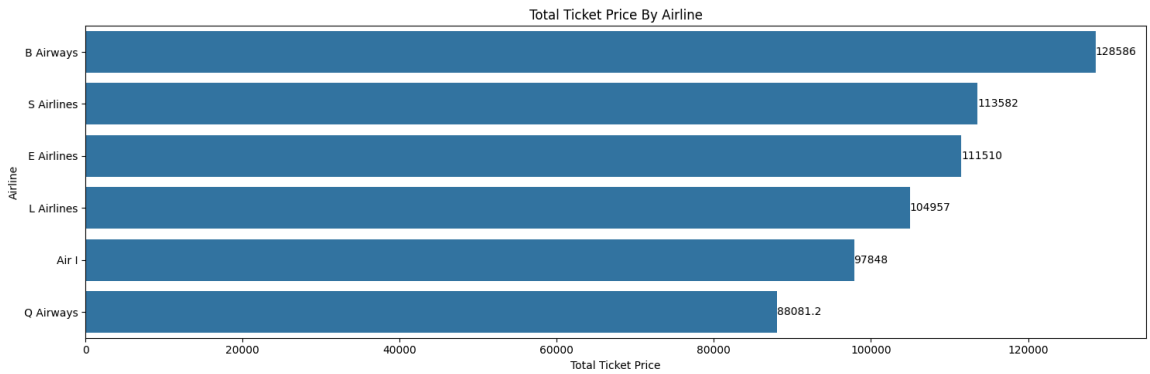
Total Ticket Price By Airline

```
Total_Ticket_Price_By_Airline = df.groupby('Airline', as_index=False)
         ['Ticket_Price'].sum().sort_values(by='Ticket_Price',ascending=
         False).reset_index(drop=True).rename(columns = {'Ticket_Price' :
         'Total_Ticket_Price'})

print(Total_Ticket_Price_By_Airline)
```

```
      Airline  Total_Ticket_Price
0   B Airways           128586.36
1  S Airlines           113582.15
2  E Airlines           111509.66
3  L Airlines           104957.42
4       Air I            97848.00
5   Q Airways            88081.17
```

```
# Visualization

plt.figure(figsize = (15, 5))
ax = sns.barplot( data = Total_Ticket_Price_By_Airline , x =
         'Total_Ticket_Price', y = 'Airline' )

for container in ax.containers :
    ax.bar_label(container)

plt.xlabel('Total Ticket Price')
```

```
plt.ylabel('Airline')
plt.title('Total Ticket Price By Airline')
plt.tight_layout()
plt.show()
```

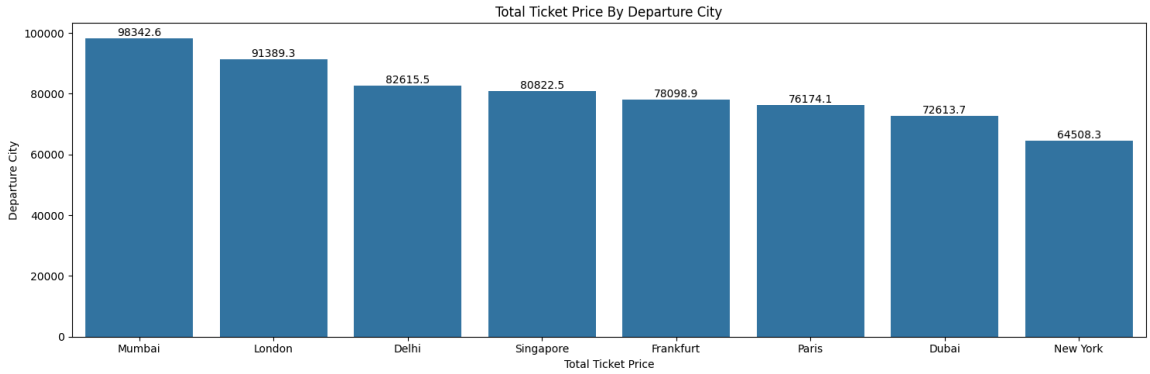


Total Ticket Price By Departure City

```
Total_Ticket_Price_By_Departure_City = df.groupby('Departure_City', as_index=
         False)['Ticket_Price'].sum().sort_values(by = 'Ticket_Price',
         ascending= False).reset_index(drop=True).rename(columns =
         {'Ticket_Price' : 'Total_Ticket_Price'})

print(Total_Ticket_Price_By_Departure_City)
```

```
  Departure_City  Total_Ticket_Price
0         Mumbai            98342.65
1         London            91389.28
2          Delhi            82615.46
3      Singapore            80822.51
4      Frankfurt            78098.85
5          Paris            76174.09
6          Dubai            72613.66
7       New York            64508.26
```

```
# Visualization

plt.figure(figsize= ( 15, 5))

ax = sns.barplot( x = 'Departure_City', y = 'Total_Ticket_Price', data =
         Total_Ticket_Price_By_Departure_City)

for i in ax.containers :
    ax.bar_label(i)

plt.title('Total Ticket Price By Departure City')
plt.xlabel('Total Ticket Price')
plt.ylabel('Departure City')
plt.tight_layout()
plt.show()
```



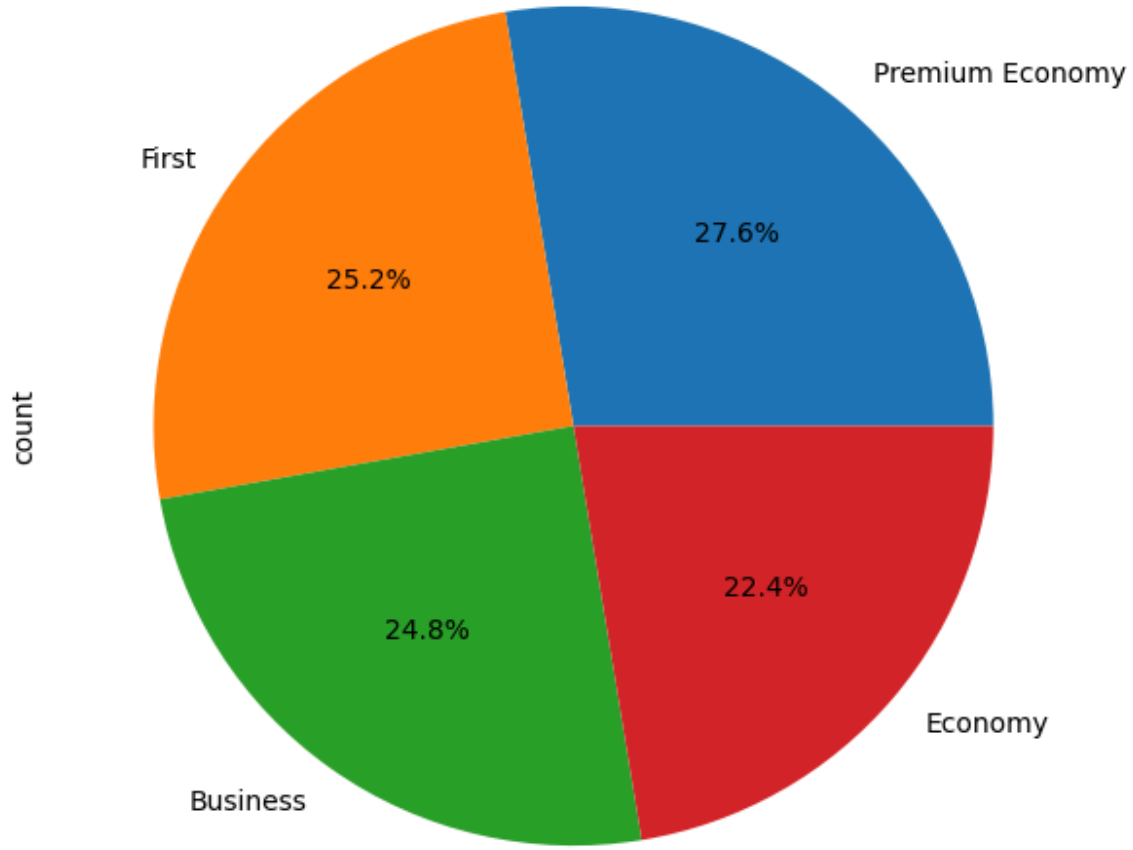
Total Bookings By Travel Class

```
Total_Bookings_By_Travel_Class = df.groupby('Travel_Class', as_index= False)
         ['Booking_ID'].nunique().sort_values(by='Booking_ID', ascending=
         False).reset_index(drop=True).rename(columns = {'Booking_ID' :
         'Total_Bookings'})

print(Total_Bookings_By_Travel_Class)
```

```
      Travel_Class  Total_Bookings
0  Premium Economy              69
1            First              63
2         Business              62
3          Economy              56
```

```
## Visualisation

plt.figure( figsize = ( 7,7 ))
df['Travel_Class'].value_counts().plot.pie(autopct = '%1.1f%%')
```

```
<Axes: ylabel='count'>
```



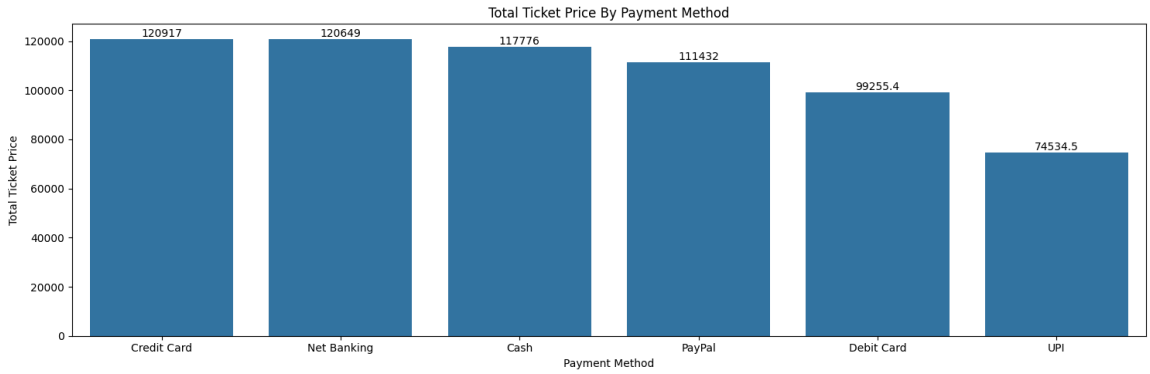
Total Ticket Price By Payment Method

```
Total_Ticket_Price_By_Payment_Method = df.groupby('Payment_Method', as_index=
         False)['Ticket_Price'].sum().sort_values(by= 'Ticket_Price',
         ascending= False).reset_index(drop=True).rename(columns =
         {'Ticket_Price' : 'Total_Ticket_Price'})

print(Total_Ticket_Price_By_Payment_Method)
```

```
  Payment_Method  Total_Ticket_Price
0    Credit Card           120917.30
1    Net Banking           120649.37
2           Cash           117776.17
3         PayPal           111431.98
4     Debit Card            99255.42
5            UPI            74534.52
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))
```

```
ax = sns.barplot( data = Total_Ticket_Price_By_Payment_Method, x =
         'Payment_Method', y = 'Total_Ticket_Price' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Total Ticket Price By Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Total Ticket Price')
plt.tight_layout()
plt.show()
```



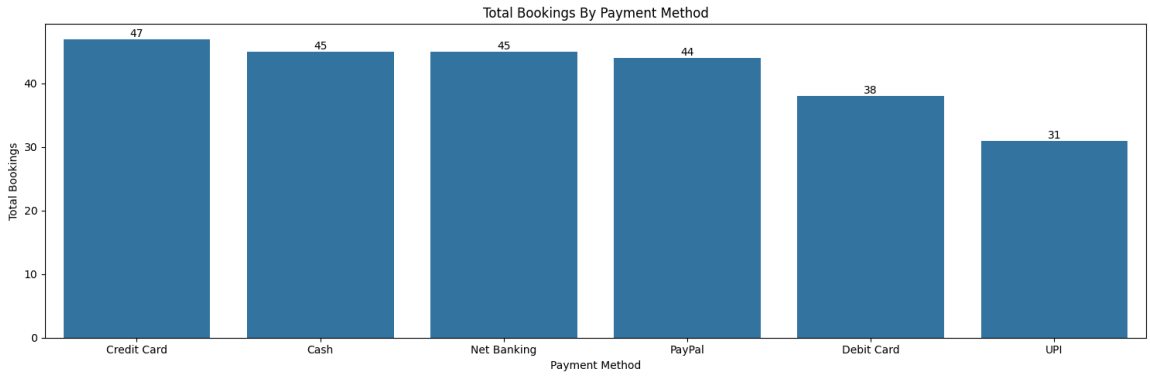
Total Bookings By Payment Method

```
Total_Bookings_By_Payment_Method = df.groupby('Payment_Method', as_index=
         False)['Booking_ID'].nunique().sort_values(by= 'Booking_ID',
         ascending= False).reset_index(drop=True).rename( columns =
         {'Booking_ID' : 'Total_Bookings'})

print(Total_Bookings_By_Payment_Method)
```

```
  Payment_Method  Total_Bookings
0    Credit Card              47
1           Cash              45
2    Net Banking              45
3         PayPal              44
4     Debit Card              38
5            UPI              31
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Total_Bookings_By_Payment_Method, x =
         'Payment_Method', y = 'Total_Bookings' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Total Bookings By Payment Method')
plt.xlabel('Payment Method')
plt.ylabel('Total Bookings')
```

```
plt.tight_layout()
plt.show()
```

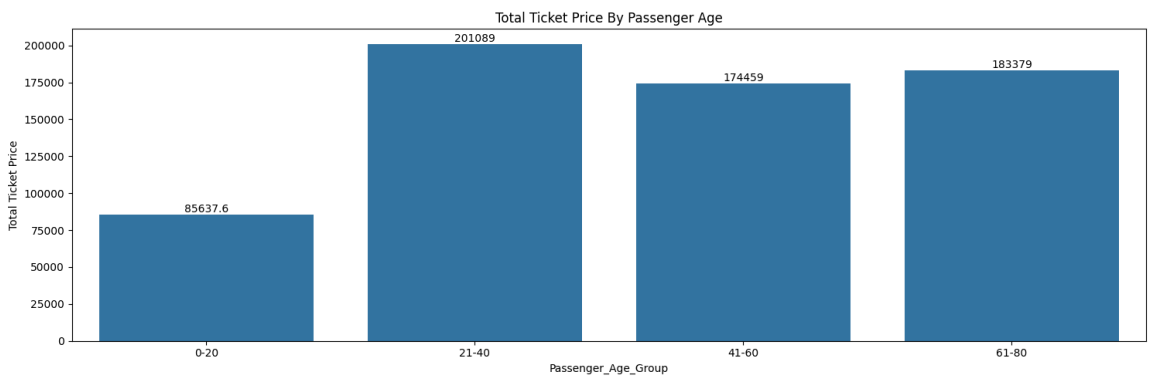


Total Ticket Price By Passenger Age

```
df['Passenger_Age'].describe()

count    250.000000
mean      45.596000
std       20.724643
min       13.000000
25%       28.000000
50%       43.000000
75%       62.000000
max       80.000000
Name: Passenger_Age, dtype: float64

## Creating Passenger Age Groups column to seggregate data into 5 age groups

bins = [ 0, 20, 40, 60, 80 ]  # age group bins
labels = [ '0-20', '21-40', '41-60', '61-80' ] # age group labels

df['Passenger_Age_Group'] = pd.cut(df['Passenger_Age'], bins = bins, labels =
         labels)
# pd.cut() --> this creates groups that is converts the numerical data into
         bins ( groups )

df['Passenger_Age_Group'] = pd.Categorical(df['Passenger_Age_Group'],
         categories=labels, ordered= False)
# pd.categorical() --> this controls order of categories or groups that helps
         in sorting

Total_Ticket_Price_By_Passenger_Age = df.groupby('Passenger_Age_Group',
         as_index= False, observed = True)
         ['Ticket_Price'].sum().sort_values(by= 'Ticket_Price',ascending=
         False).reset_index(drop = True).rename(columns = {'Ticket_Price' :
         'Total_Ticket_Price'})
# observed = True --> this shows only those categories that have data inside
         it and skips the ones with no data
# observed = False --> this shows all the categories including the ones that
         have data and the ones that dont have.

print(Total_Ticket_Price_By_Passenger_Age)
```

```
  Passenger_Age_Group  Total_Ticket_Price
0               21-40           201088.78
1               61-80           183379.23
2               41-60           174459.19
3                0-20            85637.56
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Total_Ticket_Price_By_Passenger_Age, x =
         'Passenger_Age_Group', y = 'Total_Ticket_Price' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Total Ticket Price By Passenger Age')
plt.xlabel('Passenger_Age_Group')
plt.ylabel('Total Ticket Price')
plt.tight_layout()
plt.show()
```



Flight Duration Consistency

```
Variation_In_Flight_Duration = df['Flight_Duration_hr'].std(ddof =
         0).round(2)
## .std(ddof = 0) --> this means standard deviation of whole population ;
         ddof = 1 --> sample population

print(Variation_In_Flight_Duration)
```

```
6.38
```

Total Flight Duration

```
Total_Flight_Duration = df['Flight_Duration_hr'].sum().round(2)

print(Total_Flight_Duration)
```

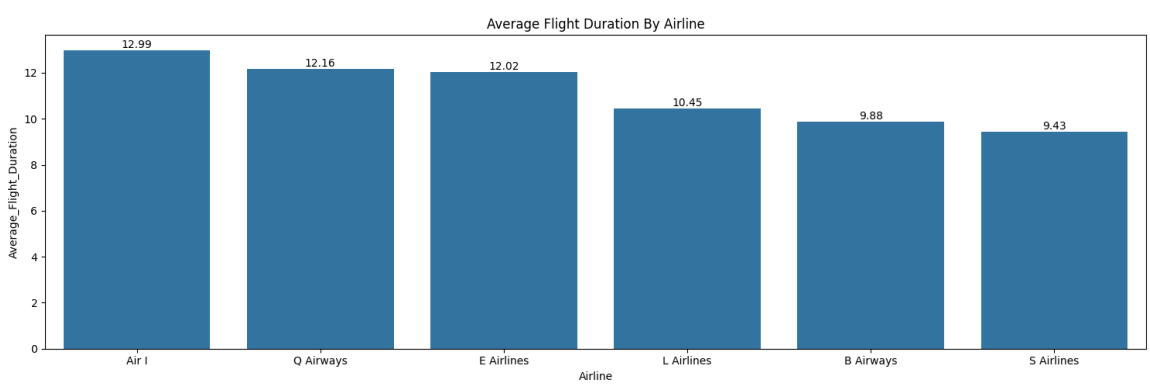
```
2760.4
```

Average Flight Duration By Airline

```
Average_Flight_Duration_By_Airline = df.groupby('Airline', as_index= False)
         ['Flight_Duration_hr'].mean().sort_values(by='Flight_Duration_hr',
         ascending= False).reset_index(drop = True).round(2).rename(columns
         = {'Flight_Duration_hr' : 'Average_Flight_Duration'})

print(Average_Flight_Duration_By_Airline)
```

```
      Airline  Average_Flight_Duration
0       Air I                    12.99
1   Q Airways                    12.16
2  E Airlines                    12.02
3  L Airlines                    10.45
4   B Airways                     9.88
5  S Airlines                     9.43
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Average_Flight_Duration_By_Airline, x = 'Airline', y
         = 'Average_Flight_Duration' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Average Flight Duration By Airline')
plt.xlabel('Airline')
plt.ylabel('Average_Flight_Duration')
plt.tight_layout()
plt.show()
```

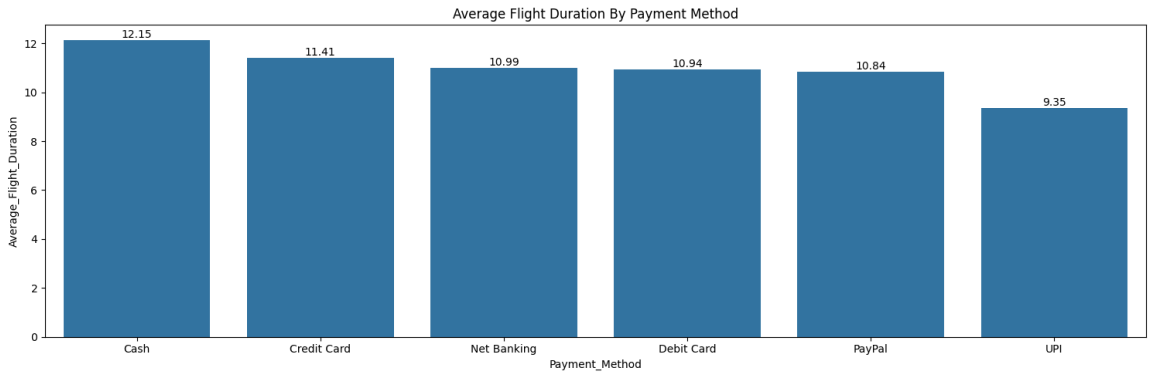


Average Flight Duration By Payment Method

```
Average_Flight_Duration_By_Payment_Method = df.groupby('Payment_Method',
         as_index= False)
         ['Flight_Duration_hr'].mean().sort_values(by='Flight_Duration_hr',
         ascending= False).reset_index(drop = True).round(2).rename(columns
         = {'Flight_Duration_hr' : 'Average_Flight_Duration'})

print(Average_Flight_Duration_By_Payment_Method)
```

```
  Payment_Method  Average_Flight_Duration
0           Cash                    12.15
1    Credit Card                    11.41
2    Net Banking                    10.99
3     Debit Card                    10.94
4         PayPal                    10.84
5            UPI                     9.35
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Average_Flight_Duration_By_Payment_Method, x =
         'Payment_Method', y = 'Average_Flight_Duration' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Average Flight Duration By Payment Method')
plt.xlabel('Payment_Method')
plt.ylabel('Average_Flight_Duration')
plt.tight_layout()
plt.show()
```



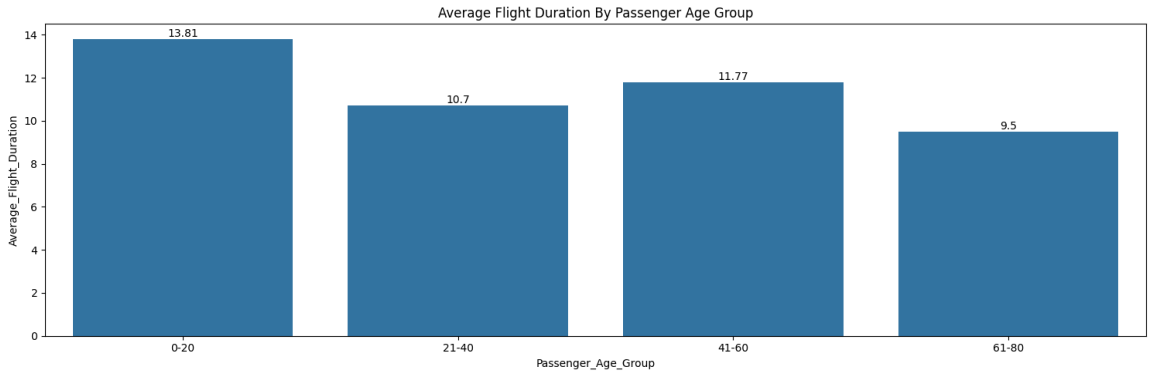
Average Flight Duration By Passenger Age

```
Average_Flight_Duration_By_Passenger_Age_Group =
         df.groupby('Passenger_Age_Group', as_index= False, observed= True)
         ['Flight_Duration_hr'].mean().sort_values(by= 'Flight_Duration_hr',
         ascending= False).reset_index(drop = True).round(2).rename(columns
         = {'Flight_Duration_hr' : 'Average_Flight_Duration'})

print(Average_Flight_Duration_By_Passenger_Age_Group)
```

```
  Passenger_Age_Group  Average_Flight_Duration
0                0-20                    13.81
1               41-60                    11.77
2               21-40                    10.70
3               61-80                     9.50
```

```
# Visualization
```

```
plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Average_Flight_Duration_By_Passenger_Age_Group, x =
         'Passenger_Age_Group', y = 'Average_Flight_Duration' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Average Flight Duration By Passenger Age Group')
plt.xlabel('Passenger_Age_Group')
plt.ylabel('Average_Flight_Duration')
plt.tight_layout()
plt.show()
```



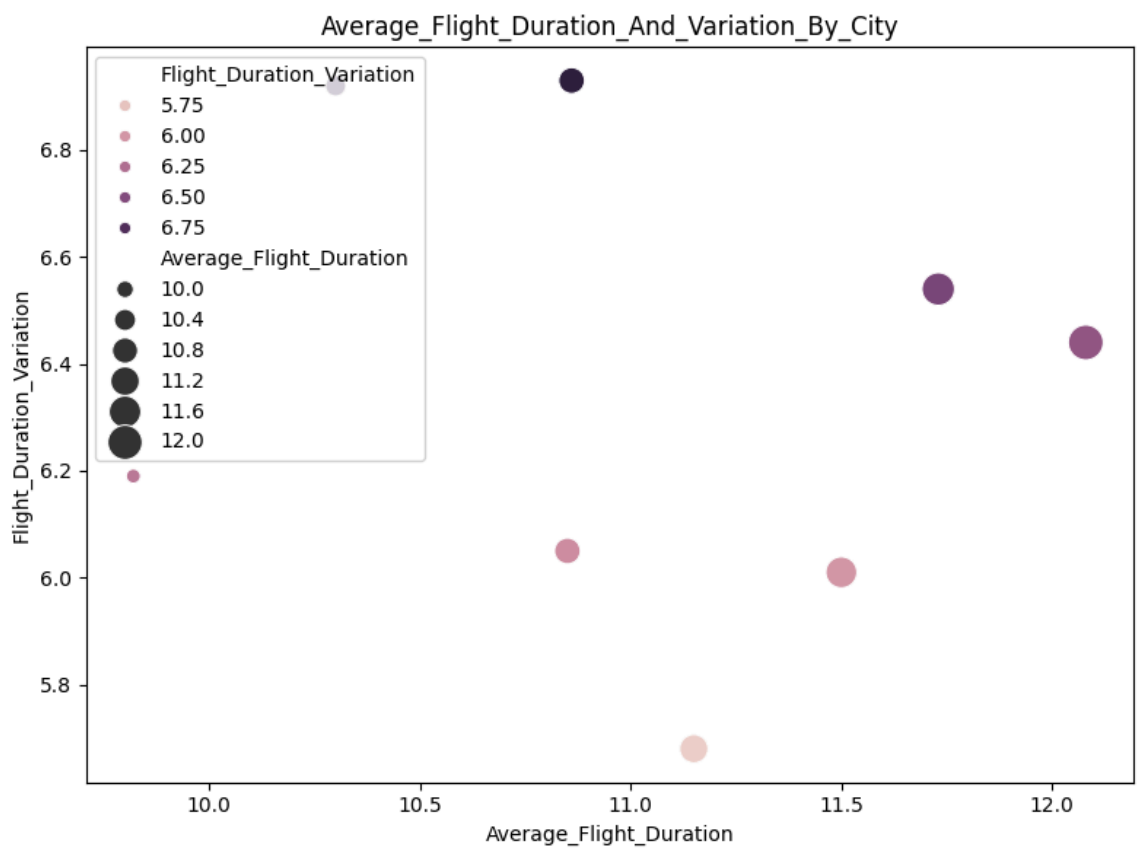
Flight Duration & Variation By City

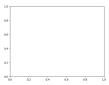
```
Average_Flight_Duration_And_Variation_By_City = df.groupby( 'Departure_City',
         as_index = False).agg( Average_Flight_Duration = (
         'Flight_Duration_hr' , 'mean' ), Flight_Duration_Variation = (
         'Flight_Duration_hr', lambda x : x.std(ddof = 0))).sort_values( by
         = [ 'Flight_Duration_Variation', 'Average_Flight_Duration' ],
         ascending = [ False, False ] ).reset_index( drop = True ).round(2)

print(Average_Flight_Duration_And_Variation_By_City)
```

```
  Departure_City  Average_Flight_Duration  Flight_Duration_Variation
0         London                   10.86                       6.93
1      Frankfurt                   10.30                       6.92
2      Singapore                   11.73                       6.54
3          Dubai                   12.08                       6.44
4       New York                    9.82                       6.19
5          Paris                   10.85                       6.05
6          Delhi                   11.50                       6.01
7         Mumbai                   11.15                       5.68
```

```
## Visualization

plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=Average_Flight_Duration_And_Variation_By_City,
    x='Average_Flight_Duration',
```

```
    y='Flight_Duration_Variation',
    size='Average_Flight_Duration',
    sizes=(50, 300),
    hue='Flight_Duration_Variation'
)

plt.title('Average_Flight_Duration_And_Variation_By_City')
plt.xlabel('Average_Flight_Duration')
plt.ylabel('Flight_Duration_Variation')

plt.tight_layout()
plt.show()

for i in range(len(Average_Flight_Duration_And_Variation_By_City)):
    plt.text(
        Average_Flight_Duration_And_Variation_By_City['Average_Flight_Duration']
         [i],
        Average_Flight_Duration_And_Variation_By_City['Flight_Duration_Variation']
         [i],
        Average_Flight_Duration_And_Variation_By_City['Departure_City'][i],
        fontsize=8
    )
```



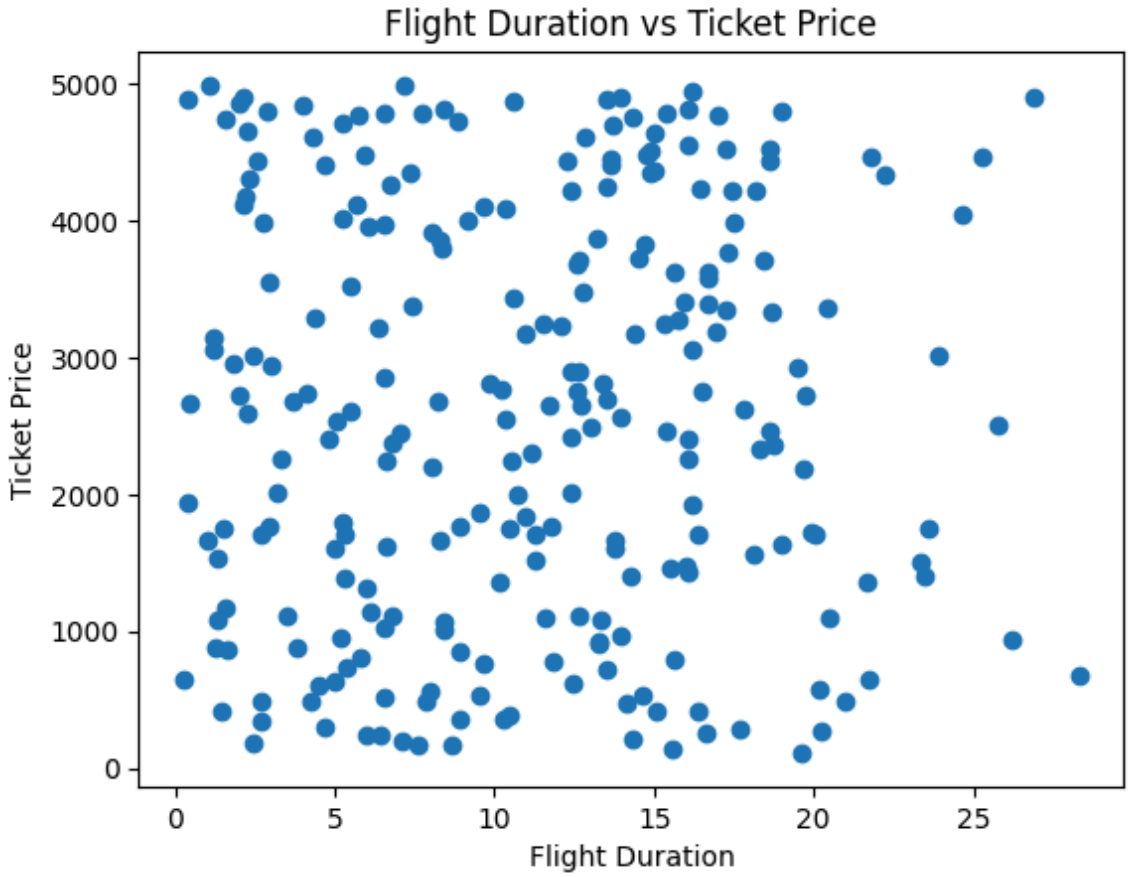


Correlation Between Flight Duration vs Ticket Price

```
Correlation = df['Flight_Duration_hr'].corr(df['Ticket_Price']).round(2)

print(Correlation)
```

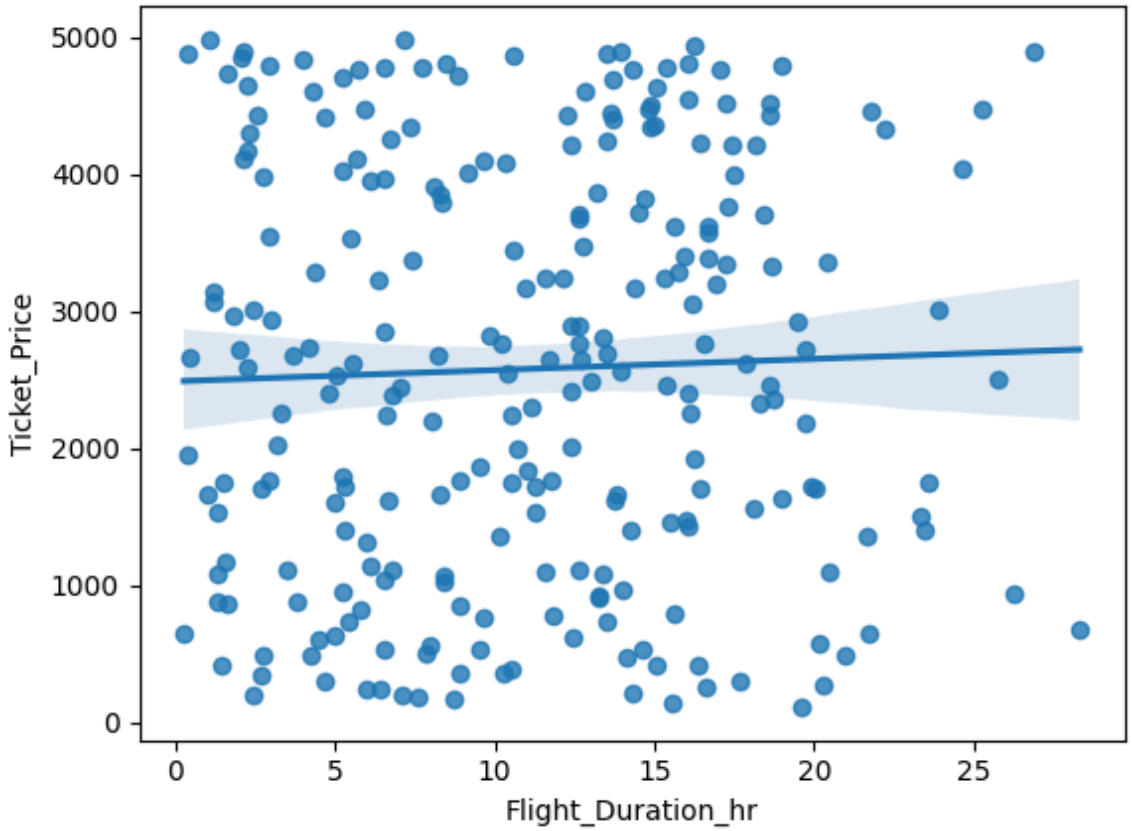
```
0.03
```

```
## Visualisation 1

plt.scatter( df['Flight_Duration_hr'], df['Ticket_Price'])
plt.xlabel('Flight Duration')
plt.ylabel('Ticket Price')
plt.title('Flight Duration vs Ticket Price')
plt.show()
```




```
## Visualization 2

sns.regplot( data = df, x = 'Flight_Duration_hr', y = 'Ticket_Price')
```

```
<Axes: xlabel='Flight_Duration_hr', ylabel='Ticket_Price'>
```



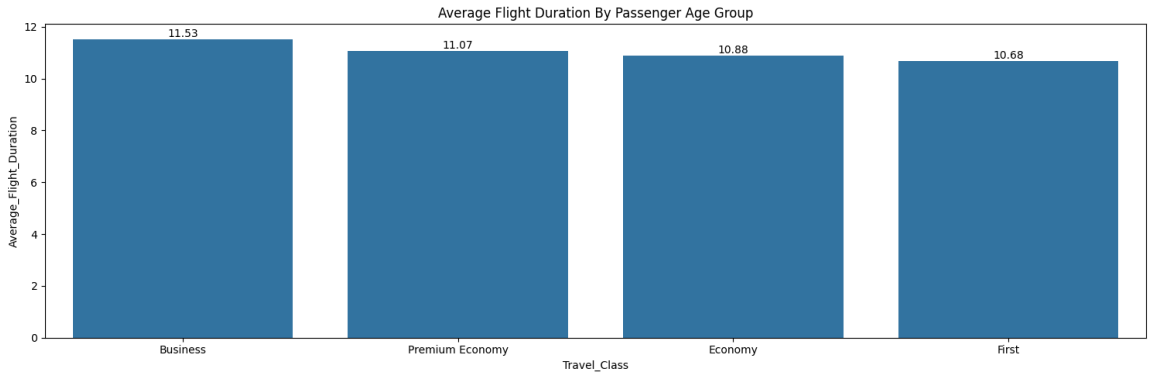
Average Flight Duration By Travel Class

```
Average_Flight_Duration_By_Travel_Class = df.groupby('Travel_Class',
         as_index= False)['Flight_Duration_hr'].mean().sort_values(by=
         'Flight_Duration_hr', ascending= False).reset_index( drop =
         True).round(2).rename(columns = {'Flight_Duration_hr' :
         'Average_Flight_Duration'})

print(Average_Flight_Duration_By_Travel_Class)
```

```
      Travel_Class  Average_Flight_Duration
0         Business                    11.53
1  Premium Economy                    11.07
2          Economy                    10.88
3            First                    10.68
```

```
# Visualization

plt.figure(figsize = ( 15, 5 ))

ax = sns.barplot( data = Average_Flight_Duration_By_Travel_Class, x =
         'Travel_Class', y = 'Average_Flight_Duration' )

for i in ax.containers :
    ax.bar_label(i)

plt.title('Average Flight Duration By Passenger Age Group')
plt.xlabel('Travel_Class')
plt.ylabel('Average_Flight_Duration')
```

```
plt.tight_layout()
plt.show()
```



Average Revenue Per Booking

```
Average_Revenue_Per_Booking = (Total_Ticket_Price / Total_Bookings).round(2)

print(Average_Revenue_Per_Booking)
```

```
2578.26
```

```
## Test to verify if one row has one booking. If both the results are same
         then it means one row has one booking
# If one row equals one booking then average ticket price equals average
         revenue per booking

df['Booking_ID'].nunique(), len(df)

# hence we can remove the kpi - Averge_Ticket_Price as they both are giving
         same results
```

```
(250, 250)
```

Average Distance Per Flight

```
Average_Distance_Per_Flight = round((Total_Distance / Total_Bookings), 2)

print(Average_Distance_Per_Flight)
```

```
7556.95
```

Total Revenue Per Km

```
Total_Revenue_Per_Km = round((Total_Ticket_Price / Total_Distance), 2)

print(Total_Revenue_Per_Km)
```

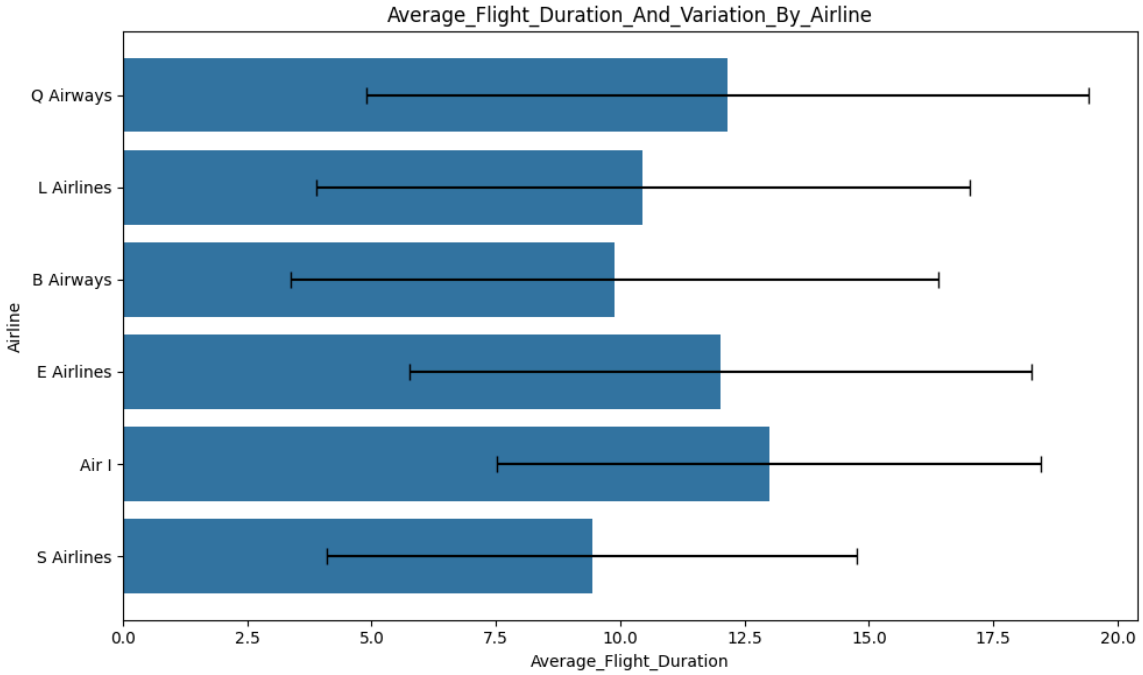
```
0.34
```

Flight Duration & Variation By Airline

```
Average_Flight_Duration_And_Variation_By_Airline = df.groupby('Airline',
         as_index= False).agg(Average_Flight_Duration = (
         'Flight_Duration_hr' , 'mean'), Flight_Duration_Variation = (
         'Flight_Duration_hr' , lambda x : x.std(ddof = 0))).sort_values( by
         = ['Flight_Duration_Variation', 'Average_Flight_Duration'],
         ascending = [ False, False ]).reset_index( drop = True ).round(2)

print(Average_Flight_Duration_And_Variation_By_Airline)
```

```
      Airline  Average_Flight_Duration  Flight_Duration_Variation
0   Q Airways                    12.16                       7.26
1  L Airlines                    10.45                       6.57
2   B Airways                     9.88                       6.51
3  E Airlines                    12.02                       6.25
4       Air I                    12.99                       5.46
5  S Airlines                     9.43                       5.32
```

```
## Visualisation

plt.figure(figsize=(10, 6))

ax = sns.barplot(
    data=Average_Flight_Duration_And_Variation_By_Airline,
    x='Average_Flight_Duration',
    y='Airline',
    orient='h'
)

plt.errorbar(
    x=Average_Flight_Duration_And_Variation_By_Airline['Average_Flight_Duration'],
    y=range(len(Average_Flight_Duration_And_Variation_By_Airline)),
    xerr=Average_Flight_Duration_And_Variation_By_Airline['Flight_Duration_Variation'],
    fmt='none',
    c='black',
    capsize=5
)

plt.title('Average_Flight_Duration_And_Variation_By_Airline')
plt.xlabel('Average_Flight_Duration')
plt.ylabel('Airline')

plt.tight_layout()
plt.show()
```


Average_Flight_Duration_And_Variation_By_Airline
Q Airways
L Airlines
B Airways
E Airlines
Air I
S Airlines
Airline
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Average_Flight_Duration